SXM GO

Gamified Island Exploration

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SPRINT 1

PRESENTATION

TEAM MEMBERS

- Oliver Skoczylas (Dev 1) – Core Auth & User Management
- Kaiden Vandegrift (Dev 2) – Map, Location & Check-In
- Taylor Knipe (Dev 3) – Gamification Engine
- Paul Han (Dev 4) – Social Features & Leaderboards

🌐 https://sxmgo.vercel.app/
💻 https://github.com/OliverSkoczylas/SXM-GO

PROJECT FOCUS

THE PROBLEM

St. Maarten has 37 beaches, 2 nations, and countless local businesses — yet tourists explore only a fraction of what the island offers.

Limited awareness and no structured discovery mechanism leaves local businesses underserved and visitors missing out.

There is no engaging, interactive way for tourists to discover hidden gems.

THE SOLUTION

- GPS-based check-ins to verify presence at 50+ locations
- Points, badges & leaderboards that reward exploration
- Compete globally and within travel groups
- Curated itineraries: Foodie, Beach Hopper, Cultural
- Drives foot traffic to local businesses

TARGET: Cruise passengers · Resort guests · Independent travelers · Local businesses seeking foot traffic

PRIORITIZED SPRINT 1 GOALS

P1

User Authentication

Email/password login, session management, user profile CRUD

Owner: Dev 1

P1

Interactive Map (50+ Locations)

Map rendering with OSM, location pins, GPS tracking, category filtering

Owner: Dev 2

P1

GPS Check-In System

50-100m radius verification, points award on check-in, duplicate prevention

Owner: Dev 2

P2

Points & Global Leaderboard

Point calculation engine, top-100 global leaderboard, real-time updates

Owner: Dev 3 / Dev 4

P2

4 Core Badges (Foodie, High Roller, Island Nomad, Sun Chaser)

Badge definitions, tracking logic, unlock notification

Owner: Dev 3

P3

Infrastructure & CI/CD Setup

Supabase backend, shared API contracts, dev environments

Owner: All Devs

MAJOR SPRINT ACHIEVEMENTS

✓ Auth System Live

COMPLETE

Email/password registration, login, JWT session management, and user profile CRUD all operational in Supabase.

Dev 1

✓ Map Rendering (10 placeholder pins)

COMPLETE

OSM integration with custom location pins, category icons, and zoom/pan gestures on both iOS & Android.

Dev 2

✓ Check-In Logic

COMPLETE

GPS-based 50m radius check-in verified; duplicate prevention within 24h; points awarded within 500ms of check-in.

Dev 2

✓ Points Engine Backend

COMPLETE

Implemented base point tracking for events and duplicate request handling, as well as tracking of challenge progress (all on DB side)

Dev 3

✓ Global Leaderboard

COMPLETE

Top-100 real-time leaderboard with Supabase Realtime; updates within 2 seconds of any point event.

Dev 4

✓ Sponsor Feedback

NOTED

Sponsors responded positively to MVP scope. Not much feedback due to non-technical background.

PROJECT BACKLOG · SPRINT BACKLOG

PROJECT BACKLOG (Selected Items)

FR-001–011	User Auth & Profiles	DONE
FR-012–026	Interactive Map & Locations	DONE
FR-027–035	🔒 GPS Check-In + Anti-Fraud	IN PROGRESS
FR-036–043	Points & Scoring System	IN PROGRESS
FR-059–071	Badges & Challenges	IN PROGRESS
FR-072–075	Leaderboard System	DONE
TR-022–026	🔒 Security (TLS, Rate Limiting, Tokens)	IN PROGRESS
FR-044–058	Itinerary System	BACKLOG
FR-084–096	Activity / Route Tracking	BACKLOG
FR-097–103	Social Features & Friends	BACKLOG

SPRINT 1 BACKLOG

✓ 🔒 Email/password auth & JWT tokens	Dev 1
✓ Supabase user profiles schema	Dev 1
✓ Google Maps integration	Dev 2
✓ 50+ location pins & category filter	Dev 2
✓ GPS check-in (50m radius)	Dev 2
○ 🔒 GPS spoofing detection (basic)	Dev 2
✓ Points engine + first-visit bonus	Dev 3
✓ 4 Badge definitions and tracking	Dev 3
✓ Global leaderboard (top 100)	Dev 4
○ 🔒 TLS/HTTPS + API rate limiting	Dev 1

🔒 = *Cybersecurity-related item*

CYBERSECURITY ASSESSMENT

Authentication Security

IMPLEMENTED

- Passwords hashed via bcrypt (industry standard)
- JWTs stored in Keychain (iOS) / Keystore (Android)
- Sessions expire after 30 days of inactivity
- Rate limiting on login endpoint (5 attempts/min)

Data Privacy

IN PROGRESS

- HTTPS/TLS 1.3 for all API communication
- Location data collected only with explicit permission
- GDPR/CCPA compliance planned (legal review pending)
- User data deletion endpoint built

GPS Anti-Fraud

PARTIAL

- Basic spoofing detection via mock location flags
- Pattern detection for impossible travel speeds
- QR fallback planned for poor GPS zones (Sprint 2)
- Manual review queue for flagged check-ins

API Security

IN PROGRESS

- API keys obfuscated in mobile bundle
- Input validation on all endpoints
- SQL injection protection via Supabase ORM
- Penetration testing scheduled for Sprint 3

WHAT WAS NOT COMPLETED · LESSONS LEARNED

NOT COMPLETED THIS SPRINT

Social OAuth (Google/Apple)

OAuth config delays — planned Sprint 2 | Dev 1

All 4 badges fully operational

Badge tracking logic needs check-in event integration | Dev 3

GPS spoofing detection (advanced)

Pattern algorithms require more data; basic checks in progress | Dev 2

Offline map caching

Deprioritized to focus on core check-in pipeline | Dev 2

TLS rate limiting on all endpoints

Auth endpoint done; remaining routes Sprint 2 | Dev 1

Push notifications

Post-MVP feature; infrastructure setup in Sprint 2 | Dev 4

LESSONS LEARNED

1. GPS Accuracy is Harder Than Expected

Urban and indoor areas degrade GPS signal. Configurable 50-100m radius and a QR fallback are both needed.

2. API Contracts Must Be Defined First

Dev 2 and Dev 3 had integration friction because check-in event payloads weren't agreed on early.

3. Supabase Free Tier Limitations

Real-time connections limited on free plan. Need to plan pricing tier before beta launch.

4. Security Cannot Be Bolted On Later

Rate limiting and token storage should be part of Sprint 1 setup, not an afterthought. Scheduling earlier next sprint.

5. Mock Endpoints Enabled Parallel Dev

Teams that used mock API endpoints were able to work independently without blocking.

TEAM CONTRIBUTIONS

Developer 1 – Oliver Skoczylas

Core Auth & User Management

Completed:

- Email/password auth system
- JWT session management
- User profile CRUD (Supabase)
- Account deletion endpoint
- GDPR data controls (partial)

Next: Social OAuth (Google/Apple)

Developer 2 – Kaiden Vandegrift

Map, Location & Check-In

Completed:

- OSM Integration (10 placeholder pins)
- Location categories
- GPS tracking
- First build of check in system
- Basic spoofing detection

Next: Offline caching, QR fallback, advanced fraud detection

Developer 3 – Taylor Knipe

Gamification Engine

Completed:

- Points calculation engine
- First-visit & streak bonuses
- 4 badge data models defined
- Duplicate request handling

Next: Frontend challenge view and check-in integration (w/ Dev 2)

Developer 4 – Paul Han

Social Features & Leaderboards

Completed:

- Global leaderboard
- Real-time Supabase updates
- Basic itinerary creation & sharing
- Leaderboard ranking backend logic

Next: Weekly/monthly boards, push notifications, itinerary builder

LIVE DEMO

Sprint 1 Feature Walkthrough

01

User Registration & Profile

02

Interactive Map (10 Locations)

03

Global Leaderboard (Real-Time)